Refusal Is Not Safety: Retrieval Gates in Health Chatbots Withhold Emergency Advice, Unevenly Across Nepali Scripts

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***Abstract*—Retrieval-augmented health chatbots often refuse when retrieved evidence appears weak. We tested whether such retrieval gates are safe for people describing emergencies and whether they behave consistently across scripts. In an evaluation pre-specified in a dated internal protocol, six gate designs were compared on 422 held-out patient messages and 100 K-QA questions, each paired across English, Devanagari Nepali, Romanized Nepali, and code-mixed text. Without a gate, the primary answer model advised urgent care for all 41 emergencies in every form. Five of the six designs left most emergencies with only a refusal, and we found no evidence that gates stopped hallucinated answers more often than helpful ones. The deployed MiniLM reranker gate refused 99% of Romanized but 1% of Devanagari messages, an effect traced to tokenization and score distributions. Normalizing queries to English restored parity across scripts, yet 56–63% of emergencies were still refused in the worst form, and a multilingual reranker gave emergency-department advice for only 44% of physician-labeled emergencies. Parity did not produce safety. Answer models also gave foreign emergency numbers in 50 of 176 English replies without a gate. Placing triage ahead of the gate (Fix-1) eliminated refusal-only replies to emergencies, although Nepali answers remained less complete than English. In a hypothesis-generating secondary analysis, a stronger pipeline (Fix-2) gave emergency-department advice for 99% of physician-labeled emergencies but sent 46% of low-acuity messages there, with higher rates in Nepali forms. Refusal in health chatbots is a clinical decision and should not be delegated to a relevance threshold.**

***Index Terms*—Abstention, clinical decision support, health chatbots, large language models, multilingual information retrieval, Nepali, retrieval-augmented generation, transliteration, triage.**

I. Introduction

HEALTH chatbots built on retrieval-augmented generation (RAG) are being deployed in settings where clinicians are scarce, including Nepal. To limit hallucinated medical advice, developers add checks between retrieval and the user: a cutoff on a reranker or embedding score, a large language model (LLM) that grades the retrieved passages, a reflection step that asks whether the evidence is sufficient, or a post-hoc checker that removes unsupported claims [1], [2]. When such a check fires, the user receives a refusal.

A refusal is usually treated as the conservative choice. For a person describing chest pain, a stroke, or an overdose, it is a missed escalation. The decision to refuse then becomes a clinical decision made by a component that was designed to judge relevance. We refer to these components as *gates*.

The problem is sharper for people who do not write in English. In a Nepali sexual and reproductive health chatbot, 61% of user queries were Romanized Nepali (Nepali written in Latin letters) and 23% were in Devanagari [3], and English-trained retrieval models score such text differently from its English equivalent [4]. A gate calibrated in English can therefore refuse a message in one script and pass the same message in another. Our motivating observation came from a deployed Nepali consumer-health chatbot, DocuMed: on the same 92 messages, its reranker gate refused 44 in English and 91 in Romanized Nepali. A stronger answer model did not change this result, because the refusal occurred before generation.

This paper asks whether this behavior is specific to one system or common to gate designs in general, and whether a design exists that avoids it. Our central claim is that *relevance gates make clinical safety decisions, and that they do so unevenly across language forms*. Equalizing a gate across scripts, we show, does not make it safe. The intended use is a consumer chatbot that answers lay people in Nepal by text, with no clinician in the loop. Our contributions are as follows.

1) A pre-specified, paired audit of six gate designs (seven variants) across four language forms and two answer models, evaluated with source labels and with five-physician acuity labels [5].
2) A mechanism for the language effect, demonstrated across the full range of cutoffs and at the tokenizer level, and evidence that parity is not safety: normalized queries and a multilingual reranker reduced script differences while gates continued to fail emergencies.
3) A decision-analytic account, which found no evidence that gates block hallucinated answers more often than helpful ones and shows that they increase expected harm under plausible costs.
4) An audit of a locale-safety hazard: foreign emergency numbers in replies intended for people in Nepal.
5) Evidence that the failure is avoidable when triage precedes the gate (Fix-1), and a secondary, hypothesis-generating analysis of a stronger screening pipeline (Fix-2) that reports its over-triage by script and separates the contributions of screener model and prompt.

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II. Related Work

A. Abstention and Refusal in Health AI

Selective prediction treats abstention as a safe action under uncertainty, but abstention can widen disparities between groups [6], and in clinical settings a reject option has a cost that should be weighed against the harm of a wrong answer [7]. Over-refusal benchmarks show safety-tuned models refusing benign requests [8], [9], and medical guardrail classifiers also block a share of benign clinical questions [10]. Health-ORSC-Bench reports that safety-optimized models refuse up to 80% of hard benign health prompts [11], and output-centric safe-completion training replaces the comply-or-refuse boundary with the most helpful safe answer [12]. Emergency triage has also been framed as a decision problem in which a model must weigh the consequences of alternative actions [13]. The refusals studied here arise in retrieval pipelines rather than in model safety tuning, and the cost we measure is missed emergency escalation.

B. Corrective and Self-Reflective RAG

Corrective RAG (CRAG) adds a retrieval evaluator that labels passages as correct, ambiguous, or incorrect and, on the latter two, turns to web search [1]. Self-RAG trains reflection tokens that decide when to retrieve and whether passages support the answer [2]. Reranker score cutoffs [14], [15] are common in practical pipelines. We implement prompt-based versions of the CRAG evaluator and of a Self-RAG-style sufficiency check without web search, because a deployment restricted to a curated medical library cannot use it. Our results therefore concern these gate patterns and not the original systems. Sufficient Context shows that large models often answer incorrectly instead of abstaining when retrieved context is insufficient, and it proposes selective generation based on sufficiency signals [16]. In clinical LLMs, evidence-sufficiency prompting lowered judged unsafe overconfidence but reduced helpfulness in a model- and judge-dependent way [17]. A large expert evaluation of medical RAG found that standard retrieval often lowered factuality and completeness relative to no retrieval [18]. These studies evaluate sufficiency as a signal of answer quality. We evaluate it as a triage decision, scored by what the person is told to do.

C. Hallucination Detection

Natural-language-inference (NLI) sentence checking [19] and fact-checking models [20] verify generated text against sources. K-QA scores long-form answers against physician-written must-have statements [21]. Its authors report that augmented retrieval reduced hallucinations, a finding that our results qualify (Section VIII).

D. Triage Benchmarks, Scripts, and Nepali

Symptom-checker audits established vignette-based triage evaluation [22]. PMR-Bench provides Reddit patient messages with urgency labels derived by GPT-5 from verified clinicians' replies [23], and AcuityBench re-rated those posts with five emergency physicians on a four-level acuity scale [5]. Ramaswamy et al. report under- and over-triage by a consumer chatbot on clinical vignettes [24], and evaluation formats can inflate apparent failure rates [25]. Script Gap studies LLM triage of maternal and newborn health messages in Indian languages and Nepali, reporting large drops for Romanized text compared with native script [26]. Neural retrievers, including multilingual ones such as BGE-M3, degrade on transliterated queries [27], and Romanized-to-Devanagari transliteration improves Nepali classification [28]. Among questions from community health workers in Nigeria, expert-rated harm was several times higher in low-resource languages than in English [29]. We extend this line of work from direct triage to the retrieval and gate layers of RAG systems, including multilingual rerankers [30], [31]. Cross-lingual medical question-answering work suggests combining English and target-language evidence [32], while most health RAG studies use English data [33].

E. Judges and Evaluation Validity

HealthBench evaluates health conversations with rubrics written by physicians [34], whereas much multilingual evaluation relies on translated benchmarks that can miss the linguistic and cultural nuances of real users [35]. Model scores also may not carry over to use by people: in a randomized study, members of the public using LLMs did not make better medical decisions than a control group, although the models alone scored highly [36]. We therefore report agreement between judges by script and treat refusal as a surrogate outcome.

F. Layered Guardrails

Multi-agent clinical guardrails screen query risk and missing context before generation [37]. We show why the order of these layers matters: a safety check placed after a gate never sees what the gate refuses.

III. Systems Compared

A. Retrieval and Generation Harness

Every gate uses the same retrieval front end: MedCPT dense retrieval [15] (top 15) over a library of 1,009 consumer-health documents (3,816 chunks; MedlinePlus, NHS, WHO, and the Epidemiology and Disease Control Division and Ministry of Health of Nepal), followed by cross-encoder reranking. Answers are generated from the top five passages with a fixed prompt that instructs the model to answer only from the documents and to direct the person to emergency care if the message may describe an emergency. The reply is written in the language and script of the user. All model calls are cached.

B. Gate Designs

We compare six designs in seven variants.

- *G1, reranker cutoff*: refuse if the best cross-encoder score is below a cutoff, with ms-marco MiniLM (as in the deployed application) and the MedCPT cross-encoder.
- *G2, dense cutoff*: the same rule applied to the MedCPT embedding cosine similarity.
- *G3, CRAG-style evaluator*: an LLM labels the passages; refuse on INCORRECT.

- *G4, Self-RAG-style sufficiency check*: an LLM judges whether the evidence is sufficient; refuse on INSUFFICIENT.
- *G5, NLI sentence checker*: the checker from the deployed application deletes sentences not supported by the passages.
- *G6, LLM groundedness check*: refuse the whole answer if an LLM judges it UNGROUNDED.

The G1 and G2 cutoffs were set on the development set, in English, to the English refusal rate of the deployed application (47.8%), and were applied unchanged to all forms, as a developer calibrating in English would do. LLM gates use gpt-oss-20b [38]. As an additional baseline, a multilingual cross-encoder, bge-reranker-v2-m3 [30], was calibrated by the same procedure.

C. Pipelines

Base: the raw message goes to retrieval and the gate, and a refusal ends the interaction. *Corrective* (exploratory): after a refusal, the query is rewritten as a normalized English query, retrieved again, and re-checked by the same gate, without web search.

Fix-1 is a language-invariant design with five elements: 1) triage runs first and is independent of the gate, and an emergency receives a fixed template containing the ambulance number for Nepal; 2) the gate decides on the normalized English query; 3) the answer is written in the language and script of the user; 4) a gate refusal is replaced by graded general guidance with warning signs; and 5) the post-generation checks G5 and G6 add a warning instead of deleting or refusing. Ablations remove triage, normalization, and the graded fallback in turn.

Fix-2 was designed after the first round to address its remaining weaknesses, and it contains no retrieval gate. Its script handling detects the form of the message and produces up to three views: the raw text, a Devanagari back-transliteration of Romanized or code-mixed text, and a faithful English translation (Claude Haiku 5.5). Its screener, Claude Haiku 5.5, applies the danger-sign checklist prompt of the deployed application to each view $K = 5$ times. For message m with views $v \in V(m)$, the score is

$$s(m) = \max_{v \in V(m)} \frac{1}{K} \sum_{j=1}^{K} \mathbf{1}[y_{v,j} = \text{EMERGENCY}] \tag{1}$$

where $y_{v,j}$ is the label of the jth sample. A multilingual red-flag lexicon (24 flags in English, Devanagari, and Romanized text) can raise urgency but never lower it. With the frozen thresholds, the system returns the emergency template if $s(m) \geq 0.6$, or if a lexicon flag is present and $s(m) \geq 0.2$; it returns urgent-today advice with warning signs and red-flag questions if $s(m) \geq 0.4$ or a flag is present; otherwise it returns a normal answer whose advice follows the triage level. Passages retrieved for the English view inform the answer without gating it, and sentences not supported by the documents are marked as such in the reply. The locale step corrects foreign emergency numbers, including those written in Devanagari digits. The lexicon has not yet been reviewed by clinicians.

D. Answer Models

The primary answer model of the first round is gpt-oss-120b [38], and Claude Sonnet 5.5 is the pre-specified second model. An exploratory check uses Claude Opus 5.5 directly. In the second round, every answer written after retrieval uses Claude Haiku 5.5, so that differences in those answers reflect the pipeline. Two Fix-1 components retain their original models in this comparison: its screener (gpt-oss-20b) and its graded fallback (gpt-oss-120b). Table VIII lists all models.

E. A Simple Model of Emergency Loss

For an emergency in language form ℓ, the probability of loss at the gate is

$$P(\text{lost} \mid \ell) = (1 - t)(1 - g_\ell) \tag{2}$$

where t is the share caught by triage before the gate and g_ℓ is the pass rate of the gate. Without triage before the gate, escalation is capped by g_ℓ regardless of the answer model, and any script gap in g_ℓ becomes a script gap in escalation.

IV. Data and Reference Standard

A. Messages and Questions

The held-out triage test set contains 362 PMR-Reddit patient posts (the r/AskDocs subset of PMR-Bench; its portal-message subsets were not used) [23] and 60 Ramaswamy vignettes [24]. Twenty-four vignettes without a binary label are excluded from binary outcomes. By source labels, 41 messages are emergencies (37 PMR and 4 Ramaswamy). The development set, used for all calibration and prompt work, contains 93 triage messages (27 emergencies; Levine and Semigran vignettes) and 100 MedQuAD retrieval questions. Answer quality was evaluated on 100 K-QA questions with physician-written must-have and nice-to-have statements [21]. Fig. 1 shows the flow of items.

B. Reference Standard

PMR-Reddit urgency labels were assigned by GPT-5 from verified clinicians' replies [23] and are not direct physician labels. The pre-specified analysis uses these source labels. As a sensitivity analysis, documented before it was run, we use AcuityBench labels for the same 362 posts: five board-certified emergency physicians per post rated acuity from A (home care) to D (emergency department now), summarized by the ordinal median, with disagreements reported separately [5]. Of our 37 PMR emergencies, 23 were consensus D, 4 C/D, 2 C, and 8 ambiguous; 19 of 325 PMR non-emergencies were consensus D. Source and physician labels agreed on 204 of 229 consensus posts ($\kappa = 0.59$). This yields 46 physician-confirmed emergencies (42 PMR consensus D and 4 Ramaswamy). The 15 not in the main sample were run through the identical pipeline.

C. Four Paired Language Forms

Every item exists in English (EN), Devanagari Nepali (DEV), Romanized Nepali (ROM), and code-mixed Nepali–English (MIX). Claude Sonnet 5.5 produced all translations. An automatic script check flagged 177 items for mixed-script errors, of which 128 were corrected by re-translation and 49

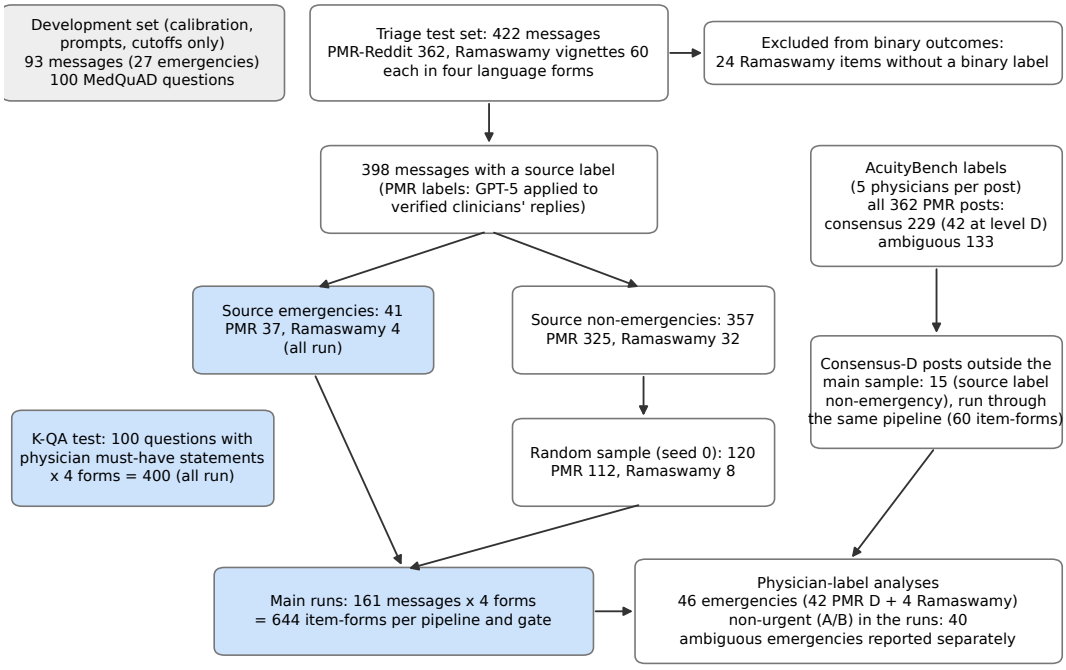


Fig. 1. Flow of items through the first-round evaluation. Shaded boxes indicate items run through every pipeline and gate.

were sent to review with a flag. Five native Nepali speakers, including the author, reviewed 782 of 795 item-forms in the first round. Eighty-one items marked for correction without a usable correction were re-translated using the note of the reviewer and reviewed again (193 of 193 item-forms). Invisible Cyrillic look-alike letters and mixed-script tokens were removed.

D. Medical-Student Labels

Three final-year medical (MBBS) students recruited for this study, separate from the AcuityBench physicians, labeled 217 messages on a four-level urgency scale, reading each message in all four language forms (277 labels, with 20 shared messages). Agreement with source labels ranged from $\kappa = 0.24$ to 0.42. Between labelers on the shared messages, Fleiss' κ was −0.04 (bootstrap 95% CI −0.28 to 0.22) and Gwet's AC1 was 0.16 (−0.12 to 0.48). Because adjudication by one labeler would have depended on which labeler rated the item, the pre-specified adjudication rule was dropped before use, and these labels are reported as a measure of ambiguity.

V. Outcomes, Hypotheses, and Statistics

A. Pre-Specified Outcomes

The first-round protocol was frozen on 27 October 2026, before any test run, as a dated internal document that was not deposited in a public registry. Its outcomes are *P1*, the share of emergencies whose final reply is a refusal with no urgent-care advice; *P2*, the share of non-emergency replies that give guidance, as a paired difference between each Nepali form and English; and *P3*, K-QA comprehensiveness (the share of must-have statements entailed; a refusal scores 0), as the same paired difference.

The second-round configuration and hypotheses were frozen on 7 November 2026, before any Fix-2 call on a test message, with SHA-256 hashes of the code and test files. Thresholds were selected on 233 tuning messages (75 emergencies; 932 item-forms) comprising the development set and AcuityBench HealthBench consensus cases that do not appear in any test set; items labeled C|D were excluded. The selection rule required recall of at least 0.95 pooled and at least 0.90 in every form, and then minimized false alarms. These test messages had been analyzed before Fix-2 was designed, so the second round is a *secondary* analysis. A confirmatory test on natively written, clinician-labeled Nepali messages is left to future work. All hypotheses and outcomes are summarized in Table I.

B. Judges and Statistics

P2 was judged by gpt-oss-20b and P3 by Qwen3.8-27b against the English statements. LLM judges saw only the reply (and, for K-QA, the question and statements), never the system that produced it. We report Wilson 95% confidence intervals for proportions, exact McNemar tests for paired binary outcomes with Holm correction across gates, and two one-sided tests (TOST, $\alpha = 0.05$) using a paired bootstrap 90% CI for equivalence.

Further analyses were added after an external review and documented before use: physician labels; a strict advice-level judge (Claude Haiku 5.5 on the reply alone, A–D or no advice); gate selectivity (risk ratio of stopping hallucinated versus helpful answers, Fisher exact test); cutoff sweeps; an emergency-number audit; and TOST power. The advice-level judge agreed with Claude Sonnet 5.5 on 83% of 120 stratified replies ($\kappa = 0.78$) and on 97% for emergency-now versus not ($\kappa = 0.92$). The help judge agreed with Claude Sonnet 5.5 on 159 of 160 replies. Back-translating 20 Nepali answers per

TABLE I
PRE-SPECIFIED HYPOTHESES AND OUTCOMES

ID	Hypothesis	Outcome
First round: gates and Fix-1		
H1	Gates turn a large share of emergencies into refusals (P1 > 0.20; lower CI bound > 0.10).	Supported for G1 (both), G2, G3, G4, and G6. G5 does not refuse.
H2	Nepali forms are refused more and lose more than English.	Supported for ROM and MIX under every gate and for DEV under LLM gates; reversed for DEV under score gates.
H3	Gates prevent some unsupported claims at a measurable cost.	No evidence of selectivity; confidence intervals were wide.
H4	Fix-1 removes suppression and makes P2 and P3 equivalent to English (TOST: ±5 points and ±0.05).	Suppression removed and P2 equivalent; P3 equivalence failed. **Not supported** as pre-specified.
Second round: Fix-2 (secondary analysis)		
V1	Fix-2 gives strict emergency-department advice at least as often as Fix-1 in every form, and for at least 90% of physician emergencies pooled.	**Passed**: 0.99 pooled vs. 0.83; Fix-2 at least as high in every form.
V2	The gap in emergency-department advice between each Nepali form and English lies within ±10 points (90% CI).	**Passed**: gaps of 0.00, +0.02, and +0.02.
V3	Over-triage of physician A/B messages is reported as a cost, with no threshold.	0.46 vs. 0.12 for Fix-1 and 0.11 with no gate.
V4	K-QA comprehensiveness gap of each Nepali form from English is within ±0.05 (TOST).	**Failed** for every Nepali form.
V5	Components are ablated from the same calls and reported beside the frozen rule.	A single screener call gave most of the gain; multi-view screening mainly added false alarms.

form changed K-QA scores by +0.010, −0.020, and +0.037 (CIs including 0), providing no evidence that the judge undercounts Nepali, at limited precision.

For the risk–coverage analysis, the expected harm per message is

$$H = \pi C m + (1 - \pi) c \tag{3}$$

where π is emergency prevalence, C is the cost of a missed emergency relative to a contradicted answer, m is the share of emergencies without urgent advice, and c is the share of K-QA answers contradicting a physician statement (a refusal contradicts nothing). The break-even ratio C^* is the value of C at which a gate and no gate have equal expected harm.

Classification of replies. P1 was assigned from the pipeline outcome: a reply counted as refusal-only when a gate ended the interaction with the fixed refusal and no urgent-care advice; template and graded replies were not refusals. The lenient escalation judge answered yes if a reply told the person to seek urgent or emergency care now, including an emergency department, an ambulance, immediate hospital care, calling 102, or seeing a doctor today. The strict advice-level judge assigned the main advice: D, emergency department or ambulance now, unconditionally; C, a doctor today or within 24–48 h; B, a doctor in the coming days or weeks; A, self-care; and N, no advice about care. It recorded separately whether the reply listed signs for seeking emergency care. Emergency-department advice in this paper means level D, which combines going to an emergency department and calling an ambulance; a general suggestion to consult a doctor is not level D. No urgent advice means level A, B, or N with no such signs. The emergency template counts as level D.

Clustered and decision analyses. Because the four forms of a message are not independent, pooled second-round estimates and contrasts use an item-clustered percentile bootstrap (5,000 resamples of messages, seed 0), and per-form tests remain paired. Net benefit [39] at emergency prevalence $\pi = 0.05$ is $\pi S - (1-\pi)F/C$, where S is the share of physician emergencies given level D, F the share of A/B messages given level D, and C the cost of a missed emergency relative to one unnecessary referral. Because level C counts as a miss, these curves use the strict definition.

VI. RESULTS: GATES AND FIX-1

A. Gates Turn Emergencies Into Refusals (H1)

Without a gate, gpt-oss-120b advised urgent care for 41 of 41 emergencies in every form. With a gate, five of six designs left most emergencies with a refusal-only reply (Fig. 2). H1 was supported for G1 (both variants), G2, G3, G4, and G6. In English alone, P1 ranged from 0.63 [0.48–0.76] for G1 MiniLM to 0.95 for G4. G5 does not refuse: it changed 72% of English answers but removed urgent-care advice from only 2% of English and 0% of Nepali emergency answers. Non-emergency help collapsed under the same gates (G4 gave help for 3 of 120 messages in English and 0 of 120 in each Nepali form; without a gate, 117–120 of 120).

The result holds with physician labels. Of 46 physician-confirmed emergencies, those receiving no urgent advice numbered 28, 0, 44, and 45 (EN, DEV, ROM, MIX) under G1 MiniLM and 44, 45, 45, and 44 under G4.

B. The Language Effect Depends on the Gate (H2)

Gate decisions differed strongly between forms of the same message ($p < 0.001$ versus English for every raw-message score gate and form; Table II). Cross-encoder gates refused nearly all Romanized and code-mixed messages (G1 MiniLM: 0.99

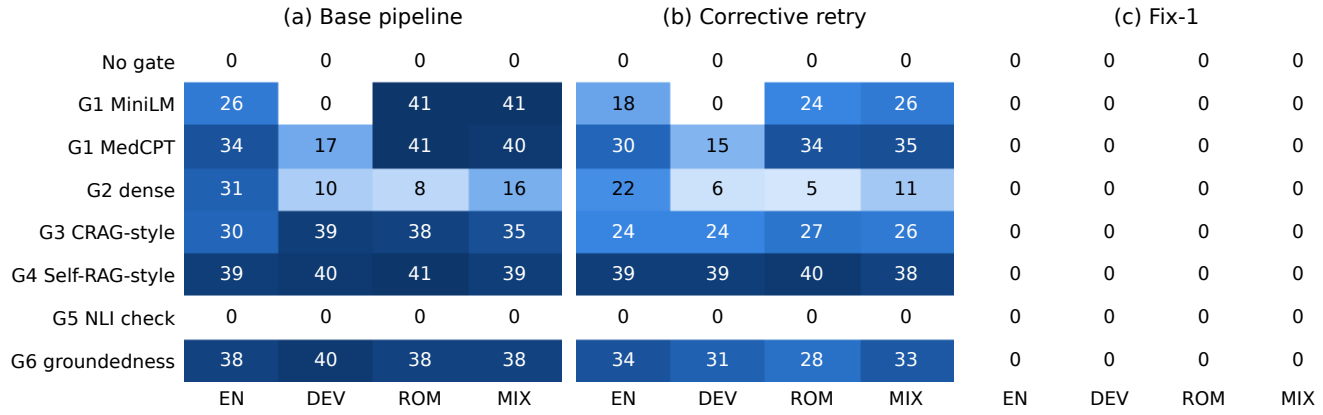


Fig. 2. Gates turned most emergencies into refusals, and Fix-1 removed these refusals. Emergencies with a refusal-only reply and no urgent-care advice, of 41 test emergencies, by gate and language form (gpt-oss-120b): (a) gate in the base pipeline, (b) with a corrective retry, and (c) with Fix-1.

versus 0.60 in English; G1 MedCPT: 0.99–1.00 versus 0.73) but diverged on Devanagari: G1 MiniLM refused almost none (0.01) and G1 MedCPT refused 0.60. LLM gates refused every Nepali form more than English (G3: 0.85–0.96 versus 0.75). H2 was therefore supported for Romanized and code-mixed text under every gate and for Devanagari under LLM gates, and reversed for Devanagari under score gates. Without any gate, Nepali answers were less complete (P3: EN 0.462, DEV 0.194, ROM 0.317, MIX 0.398).

The mechanism is visible at the tokenizer. The MedCPT tokenizer maps 85% of Devanagari tokens to the unknown token, compared with none in other forms. For G1 MiniLM, the median score was −5.0 in English, −2.7 in Devanagari, and −8.2 and −8.0 in Romanized and code-mixed text. Thus, 99% of Devanagari messages but only 1% of Romanized messages exceeded the cutoff calibrated in English: an artifact of an English-trained scorer and not a measure of relevance.

C. Parity Is Not Safety

Scoring every form on the normalized English query removed the script gap at the default operating point (refusal within 0.07 of English). Across all cutoffs (Fig. 3), no cutoff on the raw message kept the four forms within five points for either cross-encoder (0 of 19). With the normalized query most cutoffs achieved parity, but at the default operating point the gate alone still refused 56–63% of emergencies in the worst form. Parity does not make a relevance gate safe.

D. No Evidence That Gates Are Selective (H3)

Of 100 English K-QA development answers generated without a gate, 20 contradicted at least one physician statement and 66 were helpful. We found no evidence that gates stopped hallucinated answers more often than helpful ones (Table III). Random blocking would lose 3.3 helpful answers per hallucinated answer stopped, and the gates lost 2.7–3.0. The test set gave the same picture (risk ratios 0.92–2.33; no Fisher p below 0.05, lowest 0.057). Confidence intervals excluded 1 only where a gate blocked almost everything. The intervals are wide, so these data do not establish that the two rates are equal. G5 reduced contradictions from 26 to 23.

TABLE II
NORMALIZATION NARROWED SCRIPT DIFFERENCES, BUT REFUSAL STAYED HIGH (ALL 422 TEST TRIAGE MESSAGES)

Gate	Query	EN	DEV	ROM	MIX
G1 MiniLM	Raw	0.60	0.01	0.99	0.99
G1 MiniLM	Normalized	0.71	0.68	0.64	0.64
G1 MedCPT	Raw	0.73	0.60	1.00	0.99
G1 MedCPT	Normalized	0.80	0.79	0.81	0.81
G2 dense	Raw	0.63	0.19	0.18	0.27
G2 dense	Normalized	0.59	0.52	0.56	0.56

E. Harm-Weighted Risk and Coverage

At an emergency prevalence of 5%, the gate lowered expected harm only if one missed emergency cost less than $C^* = 0.5$ to 9.7 contradicted answers (one cell, G1 MiniLM on DEV, refused almost nothing and had no finite break-even). Across prevalences of 2%, 5%, and 10%, C^* ranged from 0.2 to 25.1. At $C = 10$ and 5% prevalence, the gate increased expected harm in 15 of 16 gate-by-form cells, and the same held at $C = 50$. A cutoff chosen in hindsight on the test data, which is an optimistic bound, improved on no gate by at most 0.07 in expected harm per message.

F. Corrective Retry (Exploratory)

Rewriting the query in English and retrieving again after a refusal recovered some emergencies but left most lost under G4 (38–40 of 41) and G6 (28–34). Supplement S3 gives the details.

G. Stronger Gates

The multilingual reranker bge-reranker-v2-m3, calibrated by the same procedure and with Claude Haiku 5.5 writing the answers, gave emergency-department advice for 81 of 184 physician-emergency item-forms (0.44), left 41–57% of emergencies per form without urgent advice, and withheld a reply from 42% of English and 47–60% of Nepali non-emergency messages. The script difference was far smaller than for G1 MiniLM, yet the gate remained unsafe. At 5% prevalence its break-even ratio C^* was 2.9 to 3.5. A second planned multilingual reranker (Qwen3-Reranker-0.6B [31]) could not be completed on the available CPU hardware and is not reported.

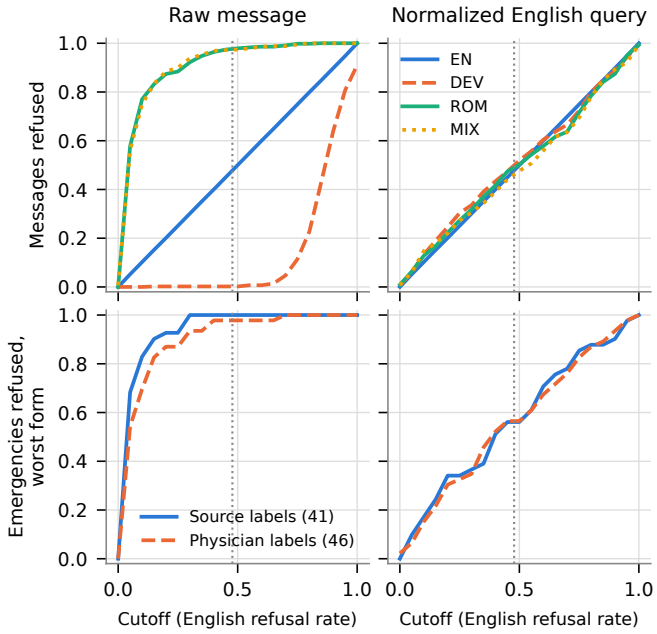


Fig. 3. No raw-message cutoff of the G1 MiniLM gate kept the four forms within five points. Top: refusal rate by language form. Bottom: emergencies refused in the worst form. The dotted line marks the operating point of the deployed application.

TABLE III
NO EVIDENCE OF SELECTIVITY: SHARE OF ANSWERS STOPPED BY EACH GATE (DEVELOPMENT K-QA, ENGLISH)

Gate	Halluc.	Helpful	Risk ratio	*p*
G1	0.35	0.29	1.22 [0.60–2.47]	0.59
G3	0.65	0.58	1.13 [0.77–1.65]	0.61
G4	0.85	0.74	1.14 [0.91–1.44]	0.38
G6	0.60	0.55	1.10 [0.72–1.67]	0.80

Two supplementary analyses, specified before they were run, tested whether a capable LLM changes this result. G7, following the idea of Sufficient Context [16], asks whether the five retrieved passages suffice to answer the raw message and refuses otherwise, with one fixed prompt and no tuning. With Claude Haiku 5.5, G7 gave emergency-department advice for 5% [2–10] of physician-emergency item-forms and left 95% [90–98] without urgent advice. With Claude Sonnet 5.5 the figures were 7% [2–13] and 93% [88–98]. No gate gave 88% [80–94] and left none. We found no evidence that G7 selectively stopped contradicted K-QA answers (Supplement S3). The failure therefore does not depend on a weak gate model.

H. Fix-1 (H4)

Compared with the same model without a gate, Fix-1 produced no refusal-only reply to any emergency under any gate, in any form, with either answer model (0 of 41 per cell; 95% upper bound 8.6%; cells reuse the same emergencies). It referred 67–96% of the 46 physician emergencies to the emergency department (no gate: 74–83%; Table IV) and left 0–1 per form without urgent advice. It gave guidance to 119–120 of 120 non-emergencies in every form, equivalent to English for every gate, and sent 5–20% of physician low-acuity posts (A/B) to the emergency department (no gate: 15–28%). It also raised Nepali answer quality above the no-gate level (DEV from 0.194 to 0.396–0.435; ROM from 0.317 to 0.371–0.457; MIX from 0.398 to 0.457–0.524; Fig. 4). These results should be read against having no gate at all, which also produced no refusal-only replies. Emergency-department advice was not consistently higher with Fix-1 (first round, 67–96% versus 74–83%; second round, pooled, 0.83 versus 0.88). The value of Fix-1 is that it keeps retrieval checks without letting them withhold a reply, and that it improved Nepali answers.

Nepali answers remained less complete than English (DEV −0.04 to −0.17; ROM −0.07 to −0.14; MIX −0.07 to +0.02). The P3 TOST failed for every gate and form, so H4 was not supported as pre-specified. The test was underpowered (power 0.00–0.17 at a true gap of zero; 250–390 questions would have been needed for 80% power), but the observed Devanagari and Romanized gaps largely exclude zero, indicating that a quality gap exists.

Ablations (G1 MiniLM). Without the graded fallback, 2–6 emergencies per form were again refusal-only and help fell to 49–65 of 120, so the fallback removes suppression. Without normalization, Devanagari quality fell from 0.435 to 0.220. Without triage, outcomes were unchanged, because the fallback and the answers carried urgent advice. This is also the main weakness of Fix-1: its screener identified only 29–35 of 41 emergencies (71–85%).

Over-escalation. Under the lenient judge (any mention of seeing a doctor today), 61–91% of non-emergency replies escalated (no gate: 59–81%), but most were advice to see a doctor soon or conditional warning signs.

I. Foreign Emergency Numbers

Answer models sometimes gave emergency numbers from other countries, a direct hazard for a person in Nepal: 50 of 176 English replies with no gate contained one. A locale rewrite removed most English instances, leaving 18, but not those inside Nepali text. Under Fix-1, 3–12 of 176 replies per form retained one.

J. Sources of the Nepali Quality Gap (Exploratory)

Answers were not truncated (Devanagari median 76 words versus 94 in English). The retrieved passages contained only 0.238 of the must-have statements, whereas English answers reached 0.464, so answers draw on model knowledge beyond the library. Writing the answer in English and translating it did not help: an English draft written from the Nepali message already scored lower (DEV 0.371, ROM 0.372, MIX 0.423 versus 0.464 for the English message), and translation lost nothing measurable. The gap therefore arises from understanding the Nepali question and not from writing Nepali. Answering the normalized English query closed the gap for code-mixed text (+0.001), partly for Devanagari (−0.046), and not for Romanized text (−0.122 [−0.189, −0.058]), where the normalized query itself loses information.

TABLE IV
FIX-1 RESTORED URGENT ADVICE TO PHYSICIAN EMERGENCIES THAT GATES HAD LOST (46 EMERGENCIES, FIRST ROUND; EN / DEV / ROM / MIX)

Pipeline	ED now	ED or 24–48 h	None urgent
No gate	38/34/34/36	45/41/42/43	0/0/0/0
G1 MiniLM, base	15/34/1/1	17/41/2/1	28/0/44/45
G4, base	1/1/1/1	2/1/1/2	44/45/45/44
Fix-1, G1 MiniLM	34/41/39/40	45/45/40/41	0/0/1/0
Fix-1, G4	31/36/38/36	42/43/41/41	0/0/1/0

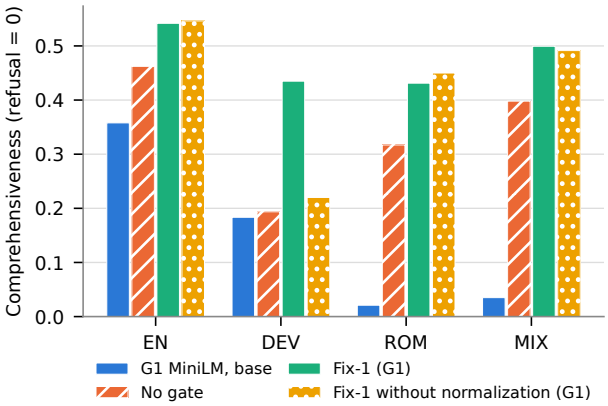


Fig. 4. Fix-1 raised Nepali answer quality, but most gaps from English remained. K-QA comprehensiveness by language form (100 questions, gpt-oss-120b).

TABLE V
FIX-2 RAISED EMERGENCY ADVICE AND OVER-TRIAGE (SECOND ROUND; EN / DEV / ROM / MIX)

System	D → ED now (of 46)	D, none urgent (of 46)	A/B → ED (of 40)
No gate	36/41/41/43	0/0/0/0	1/5/7/4
G1 MiniLM	12/41/2/1	28/0/44/45	1/4/0/0
G2 dense	11/31/28/28	33/12/14/17	1/4/6/4
bge-reranker	22/18/19/22	19/24/26/22	1/2/2/1
Fix-1	35/41/38/39	0/0/1/0	2/4/8/5
Fix-2	45/45/46/46	0/0/0/0	11/20/23/19

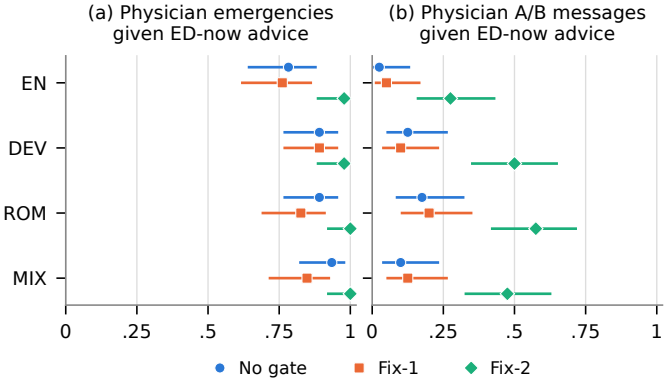


Fig. 5. Fix-2 reached nearly all physician emergencies but sent about half of low-acuity Nepali messages to the emergency department. Share given emergency-department-now advice, with Wilson 95% confidence intervals: (a) 46 emergencies and (b) 40 low-acuity (A/B) messages per form.

K. Second Answer Model

With Claude Sonnet 5.5 writing the answers, the safety results were reproduced: G6 left 34–39 of 41 emergencies with a refusal-only reply and Fix-1 left none. The P3 TOST again failed in every Nepali form. In an exploratory check (Supplement S3), Claude Opus 5.5 answering directly with no retrieval advised the emergency department for 76–80% of the 46 physician emergencies and was close to parity across forms on K-QA, but gates G1–G4 refuse before any answer model reads the message.

VII. RESULTS: FIX-2 (SECONDARY, HYPOTHESIS-GENERATING)

A. Emergencies and Over-Triage (V1–V3)

Thresholds were tuned on 233 separate messages (recall 0.950 pooled; false-alarm rate 0.258). On the test messages, Fix-2 gave strict emergency-department advice for 182 of 184 physician-emergency item-forms (0.99, item-clustered 95% CI 0.97–1.00), compared with 0.83 [0.76–0.90] for Fix-1 and 0.88 [0.80–0.94] with no gate (Table V, Fig. 5). The difference from Fix-1 was +0.16 [0.10, 0.22], so V1 passed. The two differed on 29 pairs, all favoring Fix-2. Exact McNemar tests by form gave $p = 0.002$ (EN), 0.125 (DEV), 0.008 (ROM), and 0.016 (MIX). The gaps from English were 0.00 (DEV), +0.02 (ROM), and +0.02 (MIX), with every 90% CI inside ±0.10, so V2 passed. Fix-2 left no physician emergency without urgent advice in any form.

The gain has a substantial cost (V3). Fix-2 sent 73 of 160 physician A/B item-forms (0.46 [0.33–0.58]) to emergency-department advice, compared with 0.12 [0.06–0.19] for Fix-1 and 0.11 [0.05–0.17] with no gate. Over-triage was 0.28 in English, 0.50 in Devanagari, 0.57 in Romanized, and 0.47 in code-mixed text; the gaps from English were +0.23 [0.10, 0.35], +0.30 [0.18, 0.45], and +0.20 [0.08, 0.33]. Equal detection of emergencies across scripts thus coexisted with unequal over-triage.

B. Over-Triage as an Equity Cost

The gap was not only a matter of threshold. The Fix-2 score separated physician emergencies from low-acuity messages less well in Nepali forms: AUROC was 0.873 [0.799, 0.938] in English, 0.806 [0.729, 0.882] in Devanagari, 0.756 [0.679, 0.833] in Romanized, and 0.812 [0.738, 0.887] in code-mixed text, with every interval for the difference from English excluding zero. The score takes only six values, so these estimates are coarse. Of the 73 over-triaged item-forms, 61 reached the threshold on the raw message, 11 only on a translated view, and 1 through a red-flag word.

Decision curves (Fig. 6) place these errors on one scale. At 5% emergency prevalence and $C = 20$, no gate or Fix-1 had the highest net benefit in every form. At $C = 50$, Fix-2 led only in English. A word list alone was not a viable screener (sensitivity 0.24–0.33, specificity 0.82–0.90). Whether over-triage is acceptable is a policy choice that depends on C and on the capacity of emergency services.

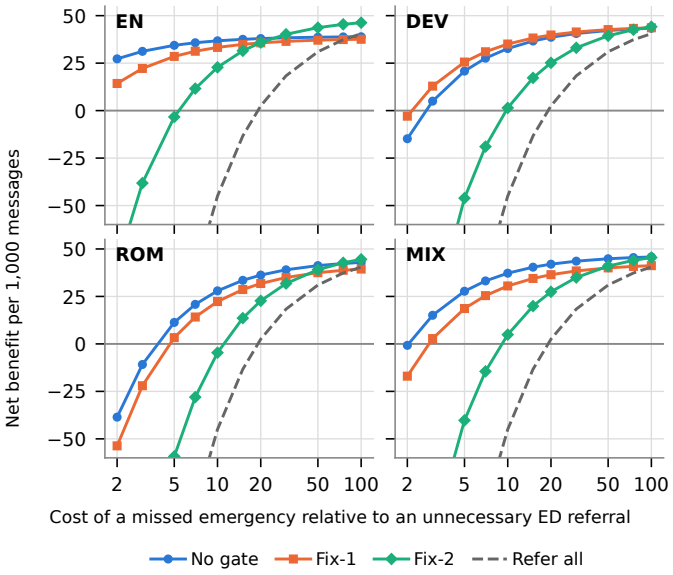


Fig. 6. At 5% emergency prevalence, Fix-2 had the highest net benefit only at high cost ratios and not in every form. The zero line is equivalent to referring no one.

C. Components and Screener Model (V5)

One call on the raw message identified 0.98 of emergencies. Adding repeated sampling changed little, and adding the other views raised recall by about one point while raising over-triage from 0.38 to 0.45. The red-flag lexicon added little (Supplement S3).

Fix-1 and Fix-2 differ in both screener model and prompt, so a supplementary analysis, specified before it was run, crossed the two on one call to the raw message (Table VI). Claude Haiku 5.5 raised recall over gpt-oss-20b by +0.19 [0.10, 0.29] with the Fix-1 prompt. The longer checklist prompt then added +0.05 [0.01, 0.10] to recall with Claude Haiku 5.5 and +0.24 [0.15, 0.33] to false alarms; with gpt-oss-20b it lowered recall by 0.07 [0.02, 0.13]. Most of the gain thus came from the screener model and most of the over-triage from the prompt. Claude Haiku 5.5 with the Fix-1 prompt reached recall of 0.93 in every form with false alarms of 0.13, which is a hypothesis for testing on unseen messages. At list prices, Fix-2 costs about $1.6 per 1,000 English messages, $2.6 per 1,000 Devanagari, and $3.8 per 1,000 Romanized or code-mixed messages, using up to 15 screening calls per message; latency was not measured.

D. Answer Quality (V4)

Fix-2 had the highest comprehensiveness of all systems in every form (Table VII), but equivalence with English within ±0.05 was not shown for any Nepali form (gaps: DEV −0.053 [−0.099, −0.005], ROM −0.130 [−0.184, −0.076], MIX −0.029 [−0.082, +0.026]), so V4 failed. The Romanized gap was larger than under Fix-1 (−0.045). The emergency template, which scores 0 on K-QA, was shown for 2 of 100 English and 8 of 100 Romanized questions and explains part of this gap. The gain must also be interpreted with its cost: Fix-2 may use model knowledge beyond the documents, whereas the no-gate comparator uses a documents-only prompt, and at least one

TABLE VI
SCREENER MODEL DROVE RECALL AND THE CHECKLIST PROMPT DROVE FALSE ALARMS (ONE CALL ON THE RAW MESSAGE; CLUSTERED 95% CI)

Model	Prompt	D → emergency	A/B → emergency
gpt-oss-20b	Fix-1	0.74 [0.64–0.84]	0.09 [0.03–0.17]
gpt-oss-20b	Checklist	0.67 [0.57–0.77]	0.10 [0.03–0.18]
Haiku 5.5	Fix-1	0.93 [0.88–0.98]	0.13 [0.06–0.22]
Haiku 5.5	Checklist	0.98 [0.95–1.00]	0.37 [0.26–0.49]

contradiction appeared in 17 of 98 scored English answers from Fix-2, compared with 8 of 100 with no gate and 17 of 100 with Fix-1.

E. Judge Agreement by Script (Supplementary)

This analysis was specified before it was run. On 160 shown replies, agreement of the advice-level judge on emergency-now versus not was 0.99 ($\kappa = 0.97$) with Claude Sonnet 5.5 and 0.96 ($\kappa = 0.91$) with Qwen3.8-27b, a judge from another model family, and it was at least 0.93 for every form. Five-level agreement with Claude Sonnet 5.5 was lowest in Devanagari (0.80, $\kappa = 0.68$). On 40 re-judged K-QA questions, Claude Sonnet 5.5 gave lower comprehensiveness than the K-QA judge by 0.060 to 0.169, with every interval excluding 0, so the values in Table VII should be compared only within that table. The Romanized gap from English was −0.080 with the K-QA judge and −0.021 with Claude Sonnet 5.5, with both intervals including 0, so its size depends on the judge. These comparisons measure agreement between models and do not replace validation by clinicians.

VIII. DISCUSSION

A. Refusal Is a Clinical Decision

Every gate we tested decides on evidence quality, and we found no evidence that any was selective for hallucinated answers. Placing such a gate in front of a health chatbot means that the most consequential decision, not answering a person in danger, is taken by a relevance threshold. Urgency should instead be decided by a clinical component that no gate can override, and evidence should shape the answer without withholding it.

B. Script Fairness Cannot Be Calibrated Into a Relevance Gate

No raw-message cutoff treated the four forms alike, and the direction of the bias depended on the model. Normalization removed the gap but not the emergency loss, and a multilingual reranker reduced the script difference while still failing most emergencies. A capable LLM sufficiency gate failed in every form. Equal treatment across scripts is therefore necessary but not sufficient for safety.

C. Sensitivity and Over-Triage Must Be Reported Together

Fix-2 shows that near-complete detection of emergencies is achievable across scripts, but at a rate of over-triage that would burden patients and emergency departments, particularly in a resource-limited health system. The screener model explained most of the gain in recall and the longer prompt most of the over-triage, so a simpler design may offer a better trade-off;

TABLE VII
FIX-2 WAS MOST COMPLETE BUT REMAINED LOWER IN NEPALI: K-QA COMPREHENSIVENESS, SECOND ROUND (CLAUDE HAIKU 5.5 ANSWERS)

System	EN	DEV	ROM	MIX
Fix-2	0.501	0.448	0.371	0.472
Fix-1	0.393	0.340	0.348	0.382
No gate	0.301	0.110	0.188	0.236
G1 MiniLM	0.252	0.103	0.018	0.059
G2 dense	0.013	0.085	0.060	0.030

this remains a hypothesis because these messages had been seen. The gap in over-triage was not only a threshold effect, and script-dependent over-triage is itself an equity concern. Studies of triage should report both errors, by script.

D. RAG Versus the Model Alone

Retrieval from our curated library did not make answers more complete: the plain model was more complete than RAG for both a large model (0.606 versus 0.461) and a small local model (0.456 versus 0.236), with no more contradictions, and the library covered only a quarter of must-have statements. This qualifies the finding that augmentation reduces hallucinations [21]: with a strong model and a limited library, grounding constrains more than it protects. The value of retrieval in this setting is auditability and local content, which supports retrieval that informs the answer and not retrieval that gates it.

E. Frontier Models

A frontier model answering directly (Supplement S3) was well calibrated and nearly equal across scripts in answer quality. This strengthens the argument: the failures we measure are produced by the system around the model. A relevance gate placed in front of the best model refuses the same emergencies, and deployments in low-resource settings often use smaller models.

IX. STRENGTHS AND LIMITATIONS

Paired items in four forms isolate the effect of script, and all gates share one retrieval front end. Outcomes and decision rules were fixed before testing, deviations were documented, two reference standards were used, and results can be regenerated from cached runs.

The study has important limitations. The test set contains 41 source-label and 46 physician-label emergencies, zero counts have upper bounds of 8–9%, and cells reuse the same items. PMR-Reddit source labels are derived from GPT-5, and physician labels cover PMR items only. Messages were translated and reviewed and were not written by Nepali speakers; they originate in English-language sources that may not reflect health-seeking or access to care in Nepal. The outcomes are surrogates: P1 does not show what a person would do after a refusal. The judges are LLMs validated against a second LLM and, for K-QA, previously against physicians; the advice-level judge has not been validated against clinicians, and clinician ratings of harm are not included in this report. The K-QA judge was more lenient than Claude Sonnet 5.5 in every form, and the size of the Romanized quality gap differed between the two judges. The medical-student labelers disagreed substantially on single messages. Gates were re-implemented from published descriptions, were prompt-based, and did not include web search. The first-round corrective, clarify-first, and frontier analyses were exploratory, and the clarify-first analysis showed that simulated patients can themselves create script gaps (Supplement S3).

The second-round evaluation is secondary because these test messages had been seen before Fix-2 was designed. Fix-2 and Fix-1 differ in screener model, prompt, and architecture; the supplementary screener test separates model from prompt, but only on messages already seen. The red-flag lexicon and Nepal emergency numbers await clinician review. K-QA analyses have limited power, and the TOST for V4 should be interpreted accordingly. External validation on natively written, clinician-labeled messages remains necessary before any conclusion about deployment.

X. CONCLUSION

Retrieval gates designed to prevent hallucination refused most emergencies, showed no evidence of selectively stopping hallucinated answers, and behaved differently across scripts. Equalizing refusal across scripts did not prevent these losses. Moving triage ahead of any gate and treating evidence as input to the answer removed refusal-only replies to emergencies. A stronger screener achieved near-complete emergency recall across scripts in a secondary analysis, at the price of frequent over-triage, which was higher in Nepali forms, and with a persistent gap in Nepali answer quality. Safety evaluations of health chatbots should treat refusal as a clinical outcome and should report missed emergencies, over-triage, and answer quality separately for each language form that users actually write.

DECLARATIONS

Ethics. The study analyzed publicly available, de-identified text and involved no human research participants, and it was not submitted for ethics committee review. The native-speaker translation reviewers and medical-student labelers contributed to the research as adult volunteers, gave written consent, and provided no personal or health information. There was no patient or public involvement in the design, conduct, or reporting of this study.

Competing interests. The author developed the DocuMed application in which the motivating observation was made and declares no other competing interests.

Funding. This work received no specific funding.

Data and code availability. No patient data were collected; Reddit posts are public and were used in the de-identified form released by PMR-Bench, and Reddit-derived text remains subject to the terms of Reddit and of PMR-Bench. Code, prompts, translations (only where source licenses permit), review data, and cached run outputs are available from the author on reasonable request and will be released publicly with the peer-reviewed version. AcuityBench labels were used for evaluation only and are not redistributed.

Use of AI tools. Claude (Anthropic) assisted with analysis code and language editing. Large language models were also the systems under study and were used for translation and as judges, as described in the Methods and the Appendix. The author designed the study, checked every result, and takes full responsibility for the content.

Supplementary material. Supplement S1 (all prompts) and Supplement S3 (supplementary results) accompany this preprint. Reporting follows TRIPOD-LLM [40] and CHART [41].

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APPENDIX

MODELS AND ACCESS

Each message was sent as a separate, stateless request, and all models were used as released without fine-tuning. Queries were made from Kathmandu, Nepal, between 28 October and 10 November 2026. Models were called by the identifiers claude-haiku-5-5, claude-sonnet-5-5, and claude-opus-5-5 (Anthropic API) and openai/gpt-oss-120b, openai/gpt-oss-20b, and qwen/qwen3.8-27b (Groq API). Each request and response was cached. Prompts were developed on the development set only by the author, a software developer who built the DocuMed application. No formal power calculation was made in advance: test sets were fixed by the available labeled data, and because of budget constraints the pre-specified scope ran all emergencies, all K-QA questions, and a random sample of 120 non-emergencies (seed 0).

TABLE VIII
MODELS USED IN THE STUDY

Role	Model and access
Answer model, first round (primary)	gpt-oss-120b, Groq API, temperature 0
Answer model, first round (second)	Claude Sonnet 5.5, Anthropic API
Frontier check	Claude Opus 5.5, Anthropic API
Answer model, second round	Claude Haiku 5.5, Anthropic API
Fix-2 screener, translation, back-transliteration	Claude Haiku 5.5, Anthropic API
Gate G7 (supplementary)	Claude Haiku 5.5 and Claude Sonnet 5.5, Anthropic API
Fix-1 screener, gates G3/G4/G6, help judge	gpt-oss-20b, Groq API, temperature 0
Fix-1 graded fallback	gpt-oss-120b, Groq API
K-QA judge	Qwen3.8-27b, Groq API, temperature 0
Advice-level judge / reference	Claude Haiku 5.5 / Claude Sonnet 5.5, Anthropic API
Retrieval, G1, G2	MedCPT encoders and cross-encoder; ms-marco-MiniLM-L-6-v2; local CPU
Multilingual gate	bge-reranker-v2-m3; local CPU
G5	nli-deberta-v3-base; local CPU

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